Curriculum Vitae — Dongkuan Zhang

Personal Information

Name	Dongkuan Zhang
Nationality	Chinese
Date of Birth	02.05.1995
Location	Dalian, Liaoning, China
Phone	+86 15510689505
Email	dkz@mail.dlut.edu.cn
Languages	Mandarin (Native), English (Fluent)

Education

2023 – Now	**Dalian University of Technology** Ph.D. in Environmental Science and Engineering (expected 2027). Supervisor: Prof. Guozhao Ji
2020 – 2023	**Minzu University of China** M.S. in Environmental Science and Engineering
2013 – 2017	**Liaoning Petrochemical University** B.S. in Petroleum Engineering

Work Experience

2017 – 2019	**Zhejiang Petrochemical Co., Ltd.** Diesel hydrocracking unit operator — equipment monitoring, process adjustment, sample collection and analysis, troubleshooting, safety management and record keeping.

Joint Training & Visiting

2025 – 2026	**Cross Laboratory, Institute of Science Tokyo** Joint PhD student supervised by Prof. Jeffrey S. Cross. Funded by the China Scholarship Council (CSC).
2026.09 – 2027.03	**University of Luxembourg** Joint PhD research visit (6 months), supervised by Prof. Bradley P. Ladewig. Funded by the DUT Global Vision Project.

Research Interests

- **Intelligent multiphase flow modeling for waste incineration**: combining physics-based CFD with AI-driven flow field prediction.

- Eulerian-Lagrangian modeling of multiphase flow, combustion and heat transfer (lab to industrial scale).
- Machine-learning and neural-network surrogates (e.g. input-convex neural networks) for fast flow prediction and non-Newtonian fluid modeling.
- Flow field design, anti-clogging and operational optimization of incinerators, circulating fluidized beds and twin-fluid nozzles.

Academic Output Snapshot

Total citations **60** · h-index **4** · i10-index **1** (updated 2026-09-01)

- Journal articles: **11** papers (incl. first/co-first author 5 papers)
- Patents: **8** patents (granted invention patents, utility models and published applications)
- Software copyrights: **2** items
- Group standards: **1** (participating drafter)
- Technology transfer: **1** patent transfer, RMB 400,000

Journal Articles (11)

2026 **Clogging mechanism and early-warning criterion for shear-thinning distillation residues in twin-fluid spray nozzles based on gas-liquid multiphase flow**

D Zhang, T Ren, Y Liu, M Li, J Yang, J Dai, J Li, JS Cross, G Ji

Chemical Engineering Research and Design · link

2026 **Numerical Investigation of Coal and Rice Husk Co-Combustion in an Industrial-Scale Circulating Fluidized Bed: Hydrodynamics, Temperature, and Pollutant Emissions**

L Liu, J Sun, YS Zhang, T Anjum, D Zhang, J Yang, J Dong, S Cheng, G Ji

Processes 14 (13), 2189 · link

2026 **Quantifying the Influence of Supercritical Drying on the Pore Size Distribution of Cobalt-Doped Silica Membrane**

T Anjum, D Zhang, G Olguin, B Ladewig, DK Wang, X Zhang, AL Khan, ...

Industrial & Engineering Chemistry Research · link

2026 **Impact of freeze drying on pore size distribution of amorphous silica membranes derived from gas permeation activation energies**

T Anjum, TH Qin, G Olguin, Y Wang, D Zhang, X Kou, DK Wang, X Zhang, ...

Separation and Purification Technology, 137736 · link · 2 citations

2025 **Input-convex neural network modeling of rheological properties of non-Newtonian fluids in chemical distillation residues**

Dongkuan Zhang, Zhiqiang Chu, Hong Tian, Guozhao Ji

Journal of Xi'an Jiaotong University, 2025, 59(12): 80-91 · link

2025	**The Eulerian–Lagrangian Model for Simulating the Moisture Content Effect on the Characteristics of MSW Combustion in a 50 T/D Grate Incinerator** J Dai, Y Du, Y Xie, D Zhang, L Liu, Y Gui, G Ji *Processes 13 (12), 3928* · link
2025	**Eulerian-Largrangian model for simulation of flow behavior and heat transfer of lab scale dual circulating fluidized beds** J Yang, D Zhang, L Liu, Y Xie, Y Du, A Li, C Qin, G Ji *Applied Thermal Engineering, 127937* · link · 5 citations
2025	**Simulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integration** D Zhang, T Anjum, Z Chu, JS Cross, G Ji *Renewable and Sustainable Energy Reviews 221, 115895* · link · 39 citations
2023	**Experimental study on sediment disturbance during deep-sea polymetallic nodule collection** Dongkuan Zhang, Meilin Liu, Jianxin Xia *Mining and Metallurgical Engineering, 2023, 43(3): 20-23* · link · 4 citations
2022	**Flow field parameter matching of deep-sea polymetallic nodule fluidization acquisition head** Dongkuan Zhang, Lei Guan, Huade Cao, et al. *Journal of Ocean Technology, 2022, 41(4): 53-60* · link · 4 citations
2021	**Comparison and analysis of deep-sea polymetallic nodule collection methods and structural parameters** Lei Guan, Dongkuan Zhang, Yuchao Xia, Jianxin Xia *Journal of Ocean Technology, 2021, 40(5): 62-70* · link · 3 citations

See the full publication list with citation counts

Patents (8)

2026	**A dual-spray-gun self-cleaning spray system** Guozhao Ji, Dongkuan Zhang, Junchao Yang, Jiacheng Dai, Hong Tian, Tong Ren *China Invention Patent Application Publication No. CN121972345A (Appl. No. CN202610181384.4, published May 5, 2026)*

2026	**An external-mixing dual-fluid atomizing spray gun** Guozhao Ji, Dongkuan Zhang *China Invention Patent Application Publication No. CN121847357A (Appl. No. CN202610181380.6, published Apr 14, 2026)*
2025	**A device for self-settling fly ash in a waste incinerator** Guozhao Ji, Dongkuan Zhang *China Utility Model Patent No. ZL 2024 2 2052228.7 (Granted CN 223137888 U)*
2024	**A device, method and application for self-settling fly ash in a waste incinerator** Guozhao Ji, Dongkuan Zhang *China Invention Patent Application Publication No. CN118912512A (Appl. No. CN202411163901.2, published Nov 8, 2024)*
2024	**A fluidized bed reactor based on Tesla Valve** Guozhao Ji, Dongkuan Zhang *China Invention Patent ZL202311155946.0 (granted; publication No. CN117504737B, Oct 1, 2024)* · Scholar
2022	**Parameter matching method for polymetallic nodule collection head** Jianxin Xia, Lei Guan, Fangfang Luan, Huade Cao, Dingbang Wei, Dongkuan Zhang *China Invention Patent Application Publication No. CN114329824A (Appl. No. CN202111566075.2)* · Scholar
2022	**A design method for flow field of polymetallic nodules collector** Jianxin Xia, Dongkuan Zhang, Fangfang Luan, Huade Cao, Dingbang Wei, Lei Guan *China Invention Patent ZL202111562069.X (granted); application publication No. CN113946927A* · Scholar
2016	**A probe for petroleum exploration** Chengchen Wu, Boyi Zhou, Shuo Ma, et al. *China Utility Model Patent No. CN201620273762.3 (granted, Aug 3, 2016)* · Scholar

Software Copyrights

2025-09	**Dual Spray Gun Self-Cleaning Control System (DSGSCS) V1.0** *China Computer Software Copyright Registration*

2025-07	**Nozzle (Pipe) Blockage Risk Monitoring Software (NRMS) V1.0** *China Computer Software Copyright Registration*

Standards

2025	**Technical Requirements for Hydrogen Production from Waste Based on High-Temperature Pyrolysis and Gasification** Dongkuan Zhang (participating drafter), et al. *Group/Industry Standard (participating author)*

Research Projects & Technology Transfer

2023 – 2024	**Self-settling fly ash technology for waste incinerators (technology transfer)** As one of the completing researchers, transferred self-developed patent achievements and related research results to industry (RMB 400,000). Personally awarded RMB 20,000 cash incentive by the university.

Honors & Awards

2026.06	Best Poster Award, The First International Symposium on Environmental Nanotechnology, Dalian, China. Poster: *Optimizing hazardous waste incineration in a full-scale rotary kiln via co-firing and staged air supply* · certificate
2026.04	DUT Global Vision Project Funding (University of Luxembourg, 6-month joint PhD visit).
2025.12	2025 Chunhui Overseas Young Excellent Achievement Collection (MOE of China). Project: "Crisis-Flow Control: Flow-field-based Hazardous Waste Incineration & Disposal Technology and Equipment."
2025.09	National Scholarship for Doctoral Students (Ministry of Education of China).
2025.11	12th Japan-China-Korea Joint Symposium on Clean Energy and Sustainable Environment, Yamanakako, Japan — Presentation Certificate.
2025.01	Drafter Certificate, Group Standard T/CITS 242-2025, China Inspection and Testing Society.

2024.12	"Environmental Engineering Star" (Dalian University of Technology).
2024.11	Award for Excellent Presentation, 2024 Korea-China-Japan Joint Symposium on Energy and Environment (Yonsei University).
2024.11	China Scholarship Council (CSC) Government-Sponsored Studentship (CSC No. 202406060182).
2024	Cash incentive for technology transfer (RMB 20,000), Dalian University of Technology.

Technical Skills

CFD	ANSYS Fluent, Star-CCM+, COMSOL Multiphysics; multiphase flow, combustion and heat transfer modeling.
Programming	Python (PyTorch, NumPy, SciPy), MATLAB; data analysis and visualization.
Experiment	Flow-field testing, heat-transfer experiments, characterization.